

Nelson-Siegel

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1 Bond Pricing Using a Nelson–Siegel Spot Curve

1.1 General pricing framework

The price P of a fixed-coupon bond with cash flows $\{C_i\}$ at times $\{t_i\}$ is given by discounting each cash flow using the corresponding spot rate:

$$P = \sum_{i=1}^N C_i D(t_i), \quad (1)$$

where $D(t)$ is the discount function. Using continuously compounded spot rates $r(t)$:

$$D(t) = e^{-r(t)t}. \quad (2)$$

Thus,

$$P = \sum_{i=1}^N C_i e^{-r(t_i)t_i}. \quad (3)$$

1.2 Nelson–Siegel spot rate specification

The Nelson–Siegel (1987) parametrization models the instantaneous spot rate as:

$$r(t) = \beta_0 + \beta_1 \frac{1 - e^{-t/\tau}}{t/\tau} + \beta_2 \left(\frac{1 - e^{-t/\tau}}{t/\tau} - e^{-t/\tau} \right). \quad (4)$$

Interpretation:

- β_0 : long-term level
- β_1 : slope (short-term component)
- β_2 : curvature (medium-term hump)
- τ : decay parameter controlling where the hump occurs

The discount function becomes:

$$D(t) = \exp \left[-t \cdot r(t) \right]. \quad (5)$$

1.3 Numerical example

Consider a 2-year annual coupon bond with:

- Face value: 100
- Coupon: 5% (so $C_1 = 5$, $C_2 = 105$)

Assume Nelson–Siegel parameters:

$$\beta_0 = 0.03, \quad \beta_1 = -0.02, \quad \beta_2 = 0.02, \quad \tau = 1.55.$$

Step 1: Compute spot rates

At $t = 1$:

$$r(1) = 0.03 - 0.02 \cdot \frac{1 - e^{-1/1.55}}{1/1.55} + 0.02 \cdot \left(\frac{1 - e^{-1/1.55}}{1/1.55} - e^{-1/1.55} \right). \quad (6)$$

Numerically:

$$r(1) \approx 0.0195.$$

At $t = 2$:

$$r(2) \approx 0.0245.$$

Step 2: Discount factors

$$D(1) = e^{-0.0195 \cdot 1} \approx 0.9807, \quad (7)$$

$$D(2) = e^{-0.0245 \cdot 2} \approx 0.9522. \quad (8)$$

Step 3: Price calculation

$$P = 5 \cdot 0.9807 + 105 \cdot 0.9522 \quad (9)$$

$$= 4.9035 + 99.9810 \quad (10)$$

$$= 104.88. \quad (11)$$

2 From model prices to yields and parameter estimation

The essential idea can be extended in two important ways. First, for any given model price P , one can compute the corresponding **yield to maturity** (YTM), denoted y , defined implicitly as the constant rate that satisfies:

$$P = \sum_{i=1}^N \frac{C_i}{(1+y)^{t_i}}. \quad (12)$$

This transformation allows model-implied prices to be expressed in terms of **model-implied yields**, which are often easier to compare with market quotations.

Second, if we observe a cross-section of bond market data (prices or yields), the Nelson–Siegel (or Svensson) parameters can be estimated by fitting the model to the data.

Let θ denote the vector of parameters (e.g. $\theta = (\beta_0, \beta_1, \beta_2, \tau)$). Two common approaches are:

Price-based estimation:

$$\min_{\theta} \sum_{j=1}^M (P_j^{\text{model}}(\theta) - P_j^{\text{market}})^2, \quad (13)$$

Yield-based estimation:

$$\min_{\theta} \sum_{j=1}^M (y_j^{\text{model}}(\theta) - y_j^{\text{market}})^2. \quad (14)$$

Here:

- $P_j^{\text{model}}(\theta)$ is obtained by discounting cash flows using the parametric spot curve
- $y_j^{\text{model}}(\theta)$ is the yield implied by that model price
- P_j^{market} and y_j^{market} are observed bond prices and yields

3 Why yield fitting and price fitting become equivalent with appropriate weighting

A useful way to understand the connection between price-based and yield-based estimation is to start from the classical first-order relationship between bond prices and yields. For a small change in yield dy , the corresponding relative change in price satisfies:

$$\frac{dP}{P} = -\text{Dur} \cdot dy, \quad (15)$$

where Dur denotes the (modified) duration of the bond. Multiplying both sides by P gives the absolute price change:

$$dP = -P \cdot \text{Dur} \cdot dy. \quad (16)$$

The quantity

$$\text{DV01} = \frac{\partial P}{\partial y} \quad (17)$$

is defined as the dollar value of a one basis point change in yield. Comparing with the expression above yields the standard identity:

$$\text{DV01} = P \cdot \text{Dur}. \quad (18)$$

Thus, price and yield errors are linked by the linear approximation:

$$\varepsilon_P \approx -\text{DV01} \cdot \varepsilon_y, \quad \varepsilon_y \approx -\frac{\varepsilon_P}{\text{DV01}}. \quad (19)$$

A yield-based least squares estimator minimizes

$$\sum_{j=1}^M \varepsilon_{y,j}^2. \quad (20)$$

Using the approximation above, this becomes

$$\sum_{j=1}^M \left(\frac{\varepsilon_{P,j}}{\text{DV01}_j} \right)^2. \quad (21)$$

This shows that yield fitting is, to first order, equivalent to price fitting with weights proportional to $1/\text{DV01}^2$. In other words, the yield-based objective function corresponds exactly to:

$$\min_{\theta} \sum_{j=1}^M \left(\frac{P_j^{\text{model}}(\theta) - P_j^{\text{market}}}{\text{DV01}_j} \right)^2. \quad (22)$$

The implication is that the two estimation approaches differ only in how they weight instruments. Long-maturity bonds have large DV01 and therefore large price sensitivity to yield changes. Yield-based estimation automatically downweights these bonds in price space, because a given yield error corresponds to a large price error. Price fitting with $1/\text{DV01}^2$ weights reproduces exactly this normalization.

Consequently, yield fitting and appropriately weighted price fitting produce nearly identical parameter estimates, especially when yield changes are small and the duration approximation is accurate.

4 Conclusions

In summary, the Nelson–Siegel framework provides a unified approach where observed bond prices or yields are used to estimate a smooth parametric spot rate curve, which in turn generates consistent model prices and model yields across maturities.